Isac S. Lainez Sevilla

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Education

University of Puerto Rico, Mayagüez Campus Expected May 2027
M.Sc. in Physics *Mayagüez, Puerto Rico*

- **GPA:** 4.00/4.00
- **Relevant Coursework:** Introduction to Theoretical Physics I–II, Theory of Electricity and Magnetism, Introduction to Quantum Theory, General Relativity, Astrophysics, and Radio Astronomy.
- **Thesis:** *Data Analysis Pipeline for Radio Pulsar Processing: Emission Studies Using Archived Observations from the Arecibo Observatory 12-m Telescope and the Green Bank Telescope*
- **Advisors:** Prof. Henri Radovan and Prof. Allison Smith

National Autonomous University of Honduras March 2019
B.S. in Physics *Tegucigalpa, Honduras*

- **Academic Distinction:** *Magna Cum Laude*
- **Final Grade:** 91/100
- **Relevant Coursework:** Classical Mechanics I–II, Quantum Mechanics I–II, Optics, Modern Physics, and Thermodynamics and Statistical Mechanics I–II.

Research Experience

Radiometer Data Analysis of Atmospheric Humidity Variability During Rainfall Events in Western Puerto Rico Jun 2025 – Aug 2025
Advisor: Prof. Hector Jimenez Gonzalez *Resilience & Action Climate Research & Its Effects of Society*

- Obtain radiometer data from the beginning of data collection through May 31, 2025.
- Divide the data by season and retain only observations collected from March 1 through September 30 of each year, corresponding to the spring and summer seasons.
- Calculate the *water vapor mixing ratio* for each observation.
- Interpolate the data using the following coordinates: time, with hourly intervals from 6 AM through 8 PM each day; and vertical atmospheric level, preferably using isobaric coordinates at 10-mb intervals from 1000 mb to 400 mb, if possible.
- Calculate the *saturation fraction* for several atmospheric layers: lower layers (850–1000 mb), lower-middle layers (600–850 mb), and upper-middle layers (400–600 mb).
- Calculate the average *saturation fraction* for each atmospheric layer and each hour of the day.
- Separate the days into rainy and dry days based on precipitation data from Mayagüez.
- Perform statistical analyses of the data, including generating plots showing the average *saturation fraction* at each hour of the day and within each atmospheric layer for rainy and dry days.

Application of Statistical Techniques to Study the Formation of Large-Scale Structure in the Universe Jan 2023 – Dec 2023

Advisor: Prof. Francisco Linares *Michoacán University of San Nicolás de Hidalgo*

- Conducted a literature review on cosmology and statistical techniques applied to cosmological analyses.
- Reproduced fundamental equations in cosmology and performed the corresponding calculations to analyze the large-scale distribution of matter.
- Developed a Python code to analyze the correlation function of galaxy convergence.
- Applied the techniques learned using synthetic data.
- Presented research findings at conferences and academic events.

Investigating the Hubble Constant H_0 Tension through Alternative Theories of General Relativity
Aug 2023 – Dec 2023
Advisor: Prof. Carlos Gabarrete *National Autonomous University of Honduras*

- Conducted a literature review of textbooks, theses, and other relevant academic sources.
- Reviewed scientific articles in the research area of interest.
- Reviewed key concepts, including the H_0 tension, modified gravity, scalar-tensor theories, the standard ΛCDM cosmological model, and statistical data analysis.
- Analyzed and compared results from different models to investigate potential explanations for or alleviation of the H_0 tension.
- Prepared the final research report.

Teaching & Mentoring

Laboratory Instructor of Physics Aug 2024 – Dec 2025
University of Puerto Rico, Mayagüez Campus

- Developed educational resources and teaching materials for practical laboratory sessions.
- Supported the professor in planning and executing laboratory and teaching activities.
- Collaborated in the development of laboratory procedures and instructional resources.
- Guided students through laboratory experiments and provided support with experimental procedures and physics concepts.
- Evaluated student laboratory work and provided feedback to support their understanding of physics concepts.

Systems Assistant Jan 2025 – May 2025
University of Puerto Rico, Mayagüez Campus

- Worked with PUPPI server infrastructure and observational data from the Arecibo Observatory.
- Diagnosed and updated server systems associated with the PUPPI project.
- Installed and configured Docker containers with NANOGrav software for undergraduate student access.
- Prepared server environments for future workshops and educational initiatives.
- Integrated and networked servers within the institutional infrastructure.

Laboratory Instructor of Physics Jan 2022 – Aug 2023

National Autonomous University of Honduras

- Developed didactic resources and teaching materials to support practical laboratory sessions.
- Assisted the professor in laboratory, teaching, research, and university-society activities.
- Participated in planning and executing teaching practices, collaborating in the development of procedures and resources, and providing guidance to students.
- Collected and analyzed laboratory data, identifying relevant statistical distributions and supporting result interpretation.

Leadership & Activities

Co-organizer, Quantum Talks in Puerto Rico Apr 2025
Mayagüez, Puerto Rico

- Developed the event program and schedule
- Assisted with event logistics and organization
- Assisted with registration and attendee coordination
- Promoted the event through social media and other outreach channels
- Moderated talks and audience discussions

Vice President & Co-Founder, Honduran Optica Chapter Jan 2023 – Jan 2024
Tegucigalpa, Honduras

- Co-founded the Honduran Optica Chapter and contributed to its establishment and development.
- Organized talks, workshops, and networking activities for students and members of the chapter.
- Facilitated communication and collaboration between students, faculty, and the broader optics and photonics community.

President, Honduran Physics Students Association (AEHFi) Jan 2023 – Jan 2024
Tegucigalpa, Honduras

- Co-founded the association and led its development as a national organization for physics students.
- Led the association and coordinated activities for physics students.
- Represented Honduran physics students in national and international academic initiatives.
- Supported the organization of student-focused academic, outreach, and networking activities.
- Facilitated communication between physics students and academic/professional communities

Delegate, The International Association of Physics Students (IAPS) Jan 2022 – Jan 2024
Tegucigalpa, Honduras

- Represented the local physics community within IAPS and contributed to international student discussions.
- Promoted IAPS activities and opportunities among physics students.
- Facilitated communication between students and the international IAPS community.

Co-organizer, Physics Career Week 2021 – 2023
Tegucigalpa, Honduras

- Developed the event program and schedule
- Assisted with event logistics and organization
- Assisted with registration and attendee coordination
- Promoted the event through social media and other outreach channels
- Moderated talks and audience discussions
- Supported communication with invited speakers
- Supported public outreach and science communication activities
- Facilitated Q&A sessions

Member, Academic Committee, VIII Central American and Caribbean Physics Olympiad 2023
Comayagua, Honduras

- Developed and prepared examination problems for the VIII Central American and Caribbean Physics Olympiad.
- Contributed to the academic design and evaluation of the competition.

Honors & Awards

Fondo Dotal Scholarship ($2,500)	2025
Academic Excellence Award, UNAH	2020, 2021 & 2023
Participant, XXIII Ibero-American Physics Olympiad	2018
First Place, Honduran Physics Olympiad	2017

Courses & Workshops

Astronomy & Physics

2026 | *Infrared Astronomy and Radio Astronomy Workshop*, NSF Arecibo Observatory, PR
2026 | *21st NRAO Synthesis Imaging Workshop*, NMT, Socorro, New Mexico
2025 | *Hands-on Training on Reactor Operations*, University of Florida, FL
2025 | *Infrared Astronomy and Radio Interferometry*, UPR–Río Piedras, PR
2025 | *ALMA Proposal*, UPR–Río Piedras, PR
2023 | *Advanced Experiments in Modern Physics*, National Autonomous University of Honduras
2021 | *Geometric Methods in Physics*, National Autonomous University of Honduras
2021 | *Introduction to the Standard Model*, National Autonomous University of Honduras
2021 | *Introduction to Cosmology*, National Autonomous University of Honduras
2021 | *Introduction to General Relativity*, National Autonomous University of Honduras

Data Science & Programming

2023 | *Data Science for Business Level 2*, IBM
2023 | *Databases and SQL for Data Science*, IBM
2022 | *Machine Learning with Python Level 1*, IBM
2022 | *Data Analysis with R Programming*, Google
2022 | *R Programming for Data Science Essentials*, IBM
2021 | *Python for Data Science*, IBM
2021 | *Data Visualization Using Python*, IBM
2021 | *Data Analysis Using Python*, IBM
2021 | *Applied Data Science with Python Level 2*, IBM
2021 | *Python Programming*, National Training Institute

Technical & Professional Tools

2023 *Introduction to Linux*, National Training Institute
2023 *Intermediate MS Excel*, National Training Institute
2021 *Adobe Illustrator Programming*, National Training Institute

Conferences and Presentations

2026 | ***Emission Properties and Variability of Millisecond Pulsars: Observations of PSR J1713+0747 with the Green Bank Telescope and PSR J0437−4715 with the Arecibo 12 m Antenna***
Authors: Isac S. Lainez Sevilla, Henri Radovan & ***Allison Smith***

Oral presentation at the Alpha-Cen Meeting and Salvadoran School of Astrophysics, San Salvador, El Salvador

2026 | ***Explorando los púlsares mediante la radioastronomía***
Authors: Isac S. Lainez Sevilla

Oral presentation at the National Autonomous University of Honduras, School of Physics, Tegucigalpa, Honduras

2026 | ***Developing a Data Analysis Pipeline for Pulsar Emission: Observations of PSR J0437−4715 and PSR B0833−45 with the Arecibo 12 m Telescope***
Authors: Isac S. Lainez Sevilla, Henri Radovan & ***Allison Smith***

Poster presentation at the 44$^{\text{th}}$ Puerto Rico Interdisciplinary Scientific Meeting, University of Puerto Rico, Río Piedras Campus

2026 | ***Single-Pulse Analysis of the Millisecond Pulsar PSR J1713+0747***
Authors: Isac S. Lainez Sevilla & ***Henri Radovan.***
Poster presentation at the 41st Interuniversity Mathematical Sciences Research Seminar, University of Puerto Rico, Río Piedras Campus

2025 | ***Pulsars: Fundamentals, Methods of Study, and Scientific Applications***
Authors: Isac S. Lainez Sevilla
Oral presentation at the University of Puerto Rico, Mayagüez Campus, Mayagüez, Puerto Rico

2024 | ***Statistical Techniques for Studying Large-Scale Structure in the Universe***
Authors: Isac S. Lainez Sevilla
Poster presentation at The Conference on Physics, Astronomy, Climate Change, and Mathematics, Copán, Honduras

2023 | ***Statistical Techniques for Studying Large-Scale Structure in the Universe***
Authors: Isac S. Lainez Sevilla
Oral presentation at the Physics Career Week 2023, Tegucigalpa, Honduras

2023 | ***Investigating the Hubble Constant (H_0) Tension through Alternative Theories of General Relativity***
Authors: Isac S. Lainez Sevilla
Oral presentation at the School of Physics Conference, Tegucigalpa, Honduras

2020 | ***Problem Solving in the III Mesoamerican Physics Olympiad***
Authors: Isac S. Lainez Sevilla
Oral presentation at the Mesoamerican Centre for Theoretical Physics (MCTP), Chiapas, Mexico

Additional Skills

Programming: Python (`numpy`, `astropy`, `scipy`, `matplotlib`, `pandas`, `seaborn`), LaTeX, R
Tools: Git/GitHub, Unix/Linux, Mathematica, PSRCHIVE, DSPSR, CARTA, CASA
Languages: Spanish (native), English (upper-intermediate), German (beginner)

References

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